

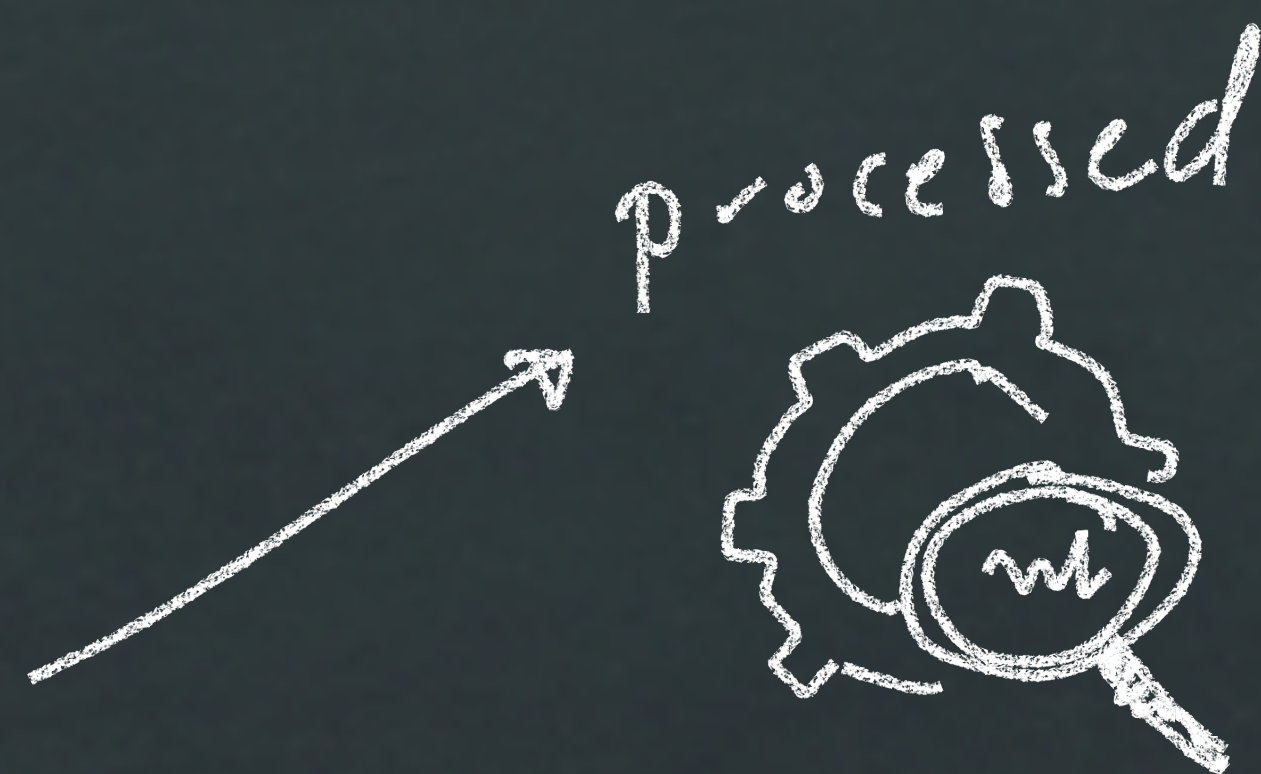
What is data?

Facts observations

figures numbers

words images

raw



information

(i)

Example of data



text
files



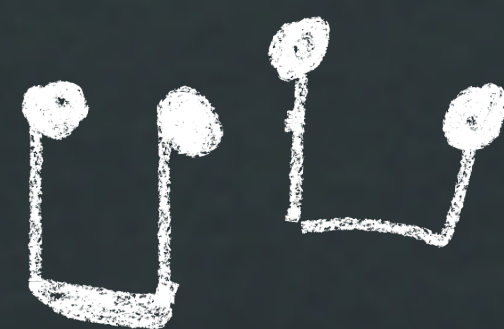
image

a	b	c
1	4	7
1	8	3
1	5	2

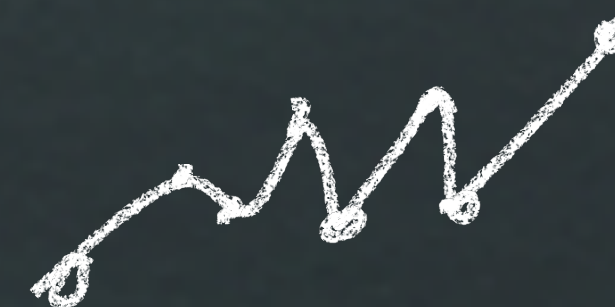
tabular
data



video



audio

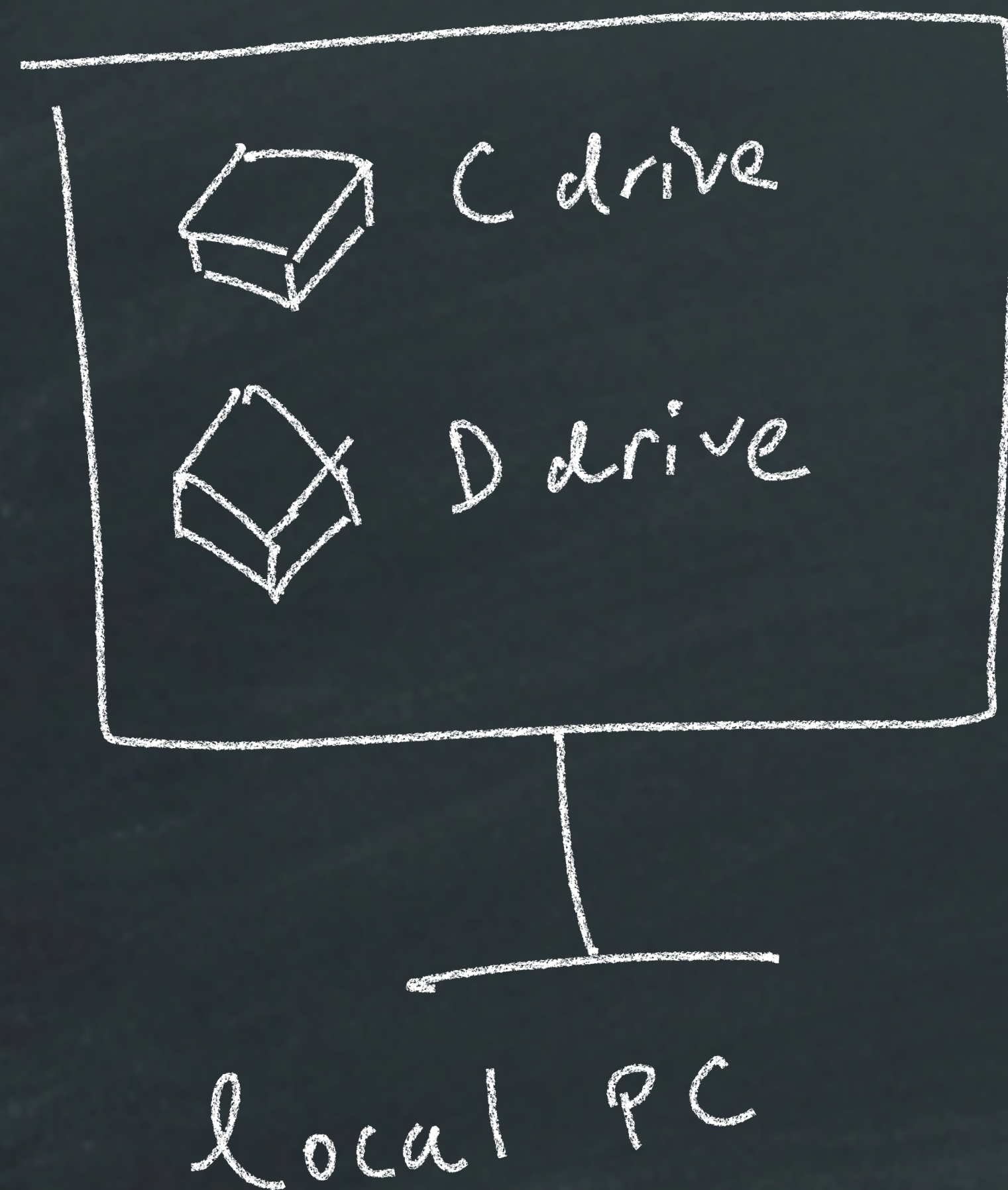


time
series

Companies & organisation generate data

data	usage
Sales	forecast revenue
customer info	improve marketing
logistic records	track delivery status
car damage image	insurance claim
technical instructions	chatbots
traffic video	identify accidents

Storing the data



Questions to address

concurrency - multiple users?

security - access to data?

scalability - run out of storage

backup & recovery - disk failure?

data redundancy - data in several places

data inconsistency - discrepancies in data

...

A database for the rescue

DBMS

ex

PostgreSQL

MySQL

Oracle DB

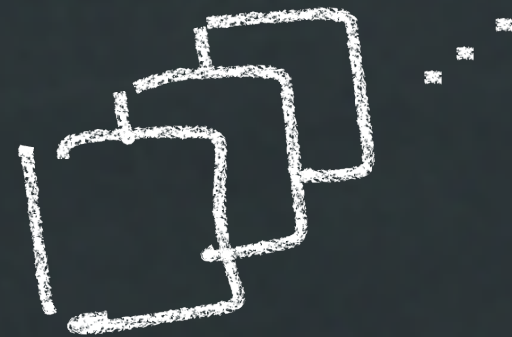
SQL server

MongoDB

Database



↓
organize
data



common
properties

concurrency control ✓

backup & recovery ✓

access control ✓

ACID transaction ✓

scalable ✓

⋮

Some real world examples

facebook → MySQL

bank of america → Oracle DB

netflix → PostgreSQL

ebay → MongoDB